Designing Data-Intensive Applications

The Big Ideas Behind Reliable, Scalable, and Maintainable Systems

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SYNOPSIS & EDITORIAL NOTE

A rigorous examination of storage engines, replication, partitioning, consensus algorithms, and batch/stream processing in modern data systems.

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CHAPTER 1 (EXCERPT)

Reliable, Scalable, and Maintainable Applications

Technical Preview Chapter

Many applications today are data-intensive, as opposed to compute-intensive. Raw CPU power is rarely the limiting factor for these applications -- bigger problems are usually the amount of data, the complexity of data, and the speed at which it is changing.

Three concerns are of recurring importance in most software systems: Reliability (the system should continue to work correctly even in the face of adversity -- hardware or software faults, and even human error), Scalability (as the system grows in data volume, traffic volume, or complexity, there should be reasonable ways of dealing with that growth), and Maintainability (over time, many different people will work on the system, and they should all be able to work on it productively).